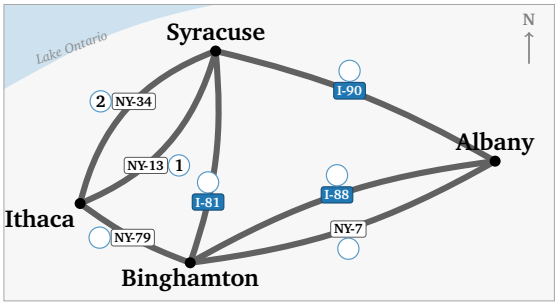
# Part 1 — Seven highways

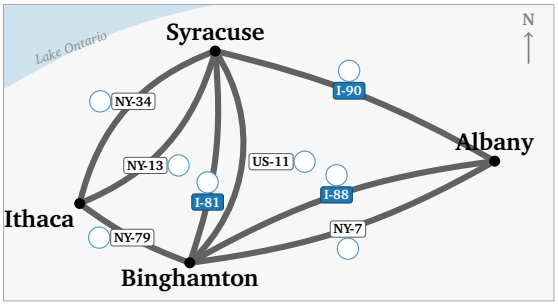
Seven highways join four Upstate New York cities: Ithaca, Syracuse, Binghamton, and Albany.

**Question 1(a)**: Drive every highway exactly once, without lifting your pencil. Start and finish anywhere; cities may repeat, highways may not. Write in each circle when you drive that road — 1 first, then 2, on to 7. Two are filled in: you start at Ithaca, take NY-13, then NY-34 straight back.

**Question 1(b)**: The same four cities, and an eighth highway: **US-11**, which really does run alongside I-81. Number your route again. Can you drive every highway without lifting your pencil?



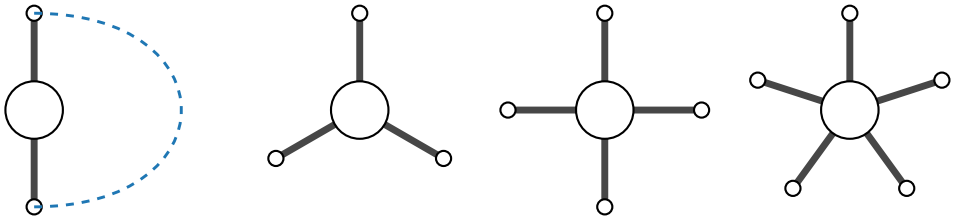

**1(a): seven highways**




**1(b): with US-11**

**Question 2**: One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**.

Join two road-ends with a curved line to mean "in here, out there". Pair up as far as you can. The first is done.



2 roads 3 roads 4 roads 5 roads

| | **2 roads** | **3 roads** | **4 roads** | **5 roads** |
|---|---|---|---|---|
| pairs you drew | 1 | | | |
| roads left over | 0 | | | |

(a) Which of these cities can you drive straight *through*?

(b) What do those numbers share?

(c) A left-over road has no partner. What must this city be for the tour?

**Question 3**: Count the highways at each city. Ithaca is done.

| | **I** | **S** | **B** | **A** | **how many have a left-over?** |
|---|---|---|---|---|---|
| seven roads | 3 | | | | |
| with US-11 too | | | | | |

One start, one finish, so at most how many cities may have a left-over road?

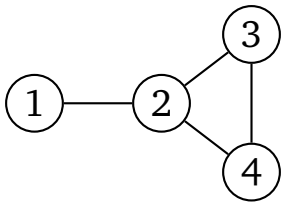
For the seven roads: can you drive every highway exactly once? If so, where does that drive begin and end?

For the eight roads, with US-11: can you drive every highway exactly once?

Why is there no route with eight?

## Part 2 — Three kinds of route

A different network: four stops, numbered 1 to 4.




**Question 4(a)**: Fill in the table. The last column takes one of these three names. First row done.

> Repeats nothing: **path**. Never repeats a line: **trail**. Repeats both: **walk**.

| route | repeats a line? | repeats a stop? | name |
|---|---|---|---|
| 1 2 3 2 4 | yes (2–3 twice) | yes (2) | |
| 1 2 3 4 2 | | | |
| 1 2 3 4 | | | |

**Question 4(b)**: Write a route from 1 to 2 that repeats a stop but no line.

**Question 4(c)**: Back to the highway map. Your drive in 1(a) used no road twice, but it did come back to a city it had already visited. Which is it — a path, a trail, or a walk?

**Question 4(d)**: It could not have been a path. How many different cities would a path of 7 roads have to touch? How many does the highway map have? Why does that rule out a path?

## Part 3 — Predictions

Part 4 hands you a notebook — a computer page that will run the network from Part 2 for you. Work the answers out on paper first; the notebook will check them.

**Question 5(a)**: Write **1** in row $i$, column $j$ if a line joins $i$ and $j$, else **0**, for the network from Part 2. Row 1 done.

| $A =$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |

**Question 5(b)**: Add up each row, then compare with the network from Part 2. What are the four row totals? What does each row total count, for that node?

**Question 5(c)**: A **2-step route** goes along one line, then along one more. From node 1 to node 3, $1 \to 2 \to 3$ is one such route. Using the network from Part 2, count the 2-step routes:

How many go from 1 to 3? Write one out.

How many go from 1 to 2? Why so few?

How many go from node 2 back to node 2?

**Question 5(d)**: The notebook in Part 4 will square your grid for you. Before it does, say what you expect to find in $A^2$.

What do you expect in row 1, column 3? In row 1, column 2?

What is on the diagonal of $A^2$, and why?

## Part 4 — The lab, on your own

**Do this one by yourself**, with this sheet beside you. The notebook takes the same four cities and seven highways you have been drawing on, and asks you to write them down in a form a computer can read.

**Point a camera at the square.** If that fails, type `go.skojaku.com/m01lab`